

Final Revision Guide

Where you stand

This version supersedes the earlier Round 2 draft. The thesis has a solid artifact, a useful evaluation question, and a quantitative benchmark whose headline values in Table 7.1 match the committed result file. The independent audit also found several issues that go beyond prose and layout, so the final pass must begin with research validity and reproducibility rather than copy-editing.

Bottom line: No supervisor can honestly guarantee a 6.0 before examination. Completing only the original checklist would not make the thesis a certain top grade because the current semantic baseline, benchmark execution, judgment provenance, statistical outputs, and user-study reporting still contain material weaknesses. Completing every gate below would make the thesis a credible top-grade candidate. Work in order: if a gate changes the results, regenerate every downstream table, figure, significance test, and claim.

Gate 1: repair the evidence before editing prose

- Step 1. **Replace the mock semantic baseline and rerun the benchmark.** In `code/ranking/ai_semantic_baseline.py`, `AISemanticBaseline()` currently defaults to `MockEmbeddingProvider`, which averages pseudo-random word vectors derived from Python's process-dependent hash function. That is neither a defensible AI semantic baseline nor fully reproducible. The recommended fix is to use a real multilingual sentence-transformer for the experiment, pin the exact model and revision, record the library versions and random seeds, and cache or checksum the embeddings. If this cannot be completed reliably, remove the semantic baseline and every comparative claim about it rather than presenting the mock model as semantic retrieval. Then regenerate Table 7.1, per-query metrics, confidence intervals, pairwise tests, win/loss counts, and all related prose.
- Step 2. **Make the benchmark genuinely frozen and describe it as a reranking experiment.** The production experiment code still calls the live CKAN `package_search` endpoint and computes recency from `datetime.now()`. All evaluated systems therefore begin from portal-generated candidates, and a future run can change when the portal or the date changes. Run the benchmark from one committed metadata snapshot with a fixed reference date. State the candidate-pool depth and make clear that the systems rerank a common portal-generated set; this design cannot measure recall for datasets that the portal candidate generator never returns. If the intended claim is end-to-end corpus retrieval, index the same frozen 500-dataset corpus independently for every system and rerun the study.
- Step 3. **Repair the relevance-judgment provenance, schema, and scale.** The public files currently tell three different stories. The interim auto/manual files contain grades 0-3, while `evaluation/ground_truth_final.json`—the file used for Table 7.1—contains grades 0-2. In `evaluation/data/pooled_candidates.csv`, `judge1_grade`, `judge2_grade`, and `adjudicated_grade` are identical for all 150 rows, so those fields do not evidence two independent assessors. Rebuild or relabel the schema to reflect the real single-author assessment and consolidation process; document any 0-3 to 0-2 mapping; and replace the misleading `adjudicated` annotator label if it does not describe an actual adjudication. The evaluation should fail if the final ground truth is missing instead of silently copying an auto-labelled fallback into the final filename.

Required metric statement: identify `ground_truth_final.json` as the authoritative file; define every final grade; state that MAP, P@5 and MRR treat grades greater than zero as relevant if that remains the implemented rule; and state that nDCG uses exponential gains of $2^{\text{grade} - 1}$. The published judgments plus this paragraph must reproduce Table 7.1.

- Step 4. **Make the user-study evidence auditable.** Appendix A lists 10 questionnaire items, but Table 7.2 reports 21. Insert the exact instrument that participants actually completed, including item wording, order, response scale, and any open-ended prompts. No anonymized response matrix is present in the repository version inspected; publish the de-identified item-level responses and the script that computes the means and standard deviations if consent permits, or document a clear audit trail and data-access restriction. Reconcile the demographic source as well. If no open-ended data were collected and analysed, stop calling this a qualitative evaluation or a mixed-method study; describe it as a formative, descriptive questionnaire combined with a quantitative retrieval benchmark. If open-ended data exist, report the coding method and actual themes rather than referring vaguely to qualitative feedback.
- Step 5. **Correct the statistical output pipeline before publishing its tables.** `evaluation/experiment_runner.py` currently discards the Wilcoxon statistic and writes 0.0 for every comparison. Store the statistic returned by SciPy, define the zero-difference handling, and handle identical vectors explicitly instead of leaving an unexplained NaN. The confidence intervals in `system_summary.csv` are t-intervals, not the bootstrap intervals promised in Chapter 6, and several extend above 1.0 for bounded metrics. Use one defensible method, label it accurately, and regenerate the files. Under the current results, fuzzy-versus-semantic differences do not survive Holm correction, so describe only higher point estimates unless the corrected rerun supports a stronger statement.

Gate 2: align the questions, claims, and implemented system

- Step 6. **Use one set of research questions everywhere.** The RQ1-RQ4 headings in Section 7.8 are different from the questions in Chapter 1 and Section 8.3, while the README contains a third set. Choose one evidence-compatible set and repeat the wording verbatim in the Introduction, methodology, Chapter 7 discussion, Chapter 8, abstract, and repository. Pay particular attention to RQ1's word 'improve': the proposed system does not outperform the portal or metadata baseline on the aggregate metrics. RQ4's word 'improving' is also causal, but a single-condition self-report study cannot establish causal improvement. Either revise these questions consistently or answer them explicitly with the negative/limited evidence the study produced.
- Step 7. **Correct the false numerical and feature claims.** Section 7.8 says the proposed system achieved the highest MRR. Table 7.1 shows Portal and Metadata at 0.9000 and Fuzzy at 0.8333. Correct this everywhere. Chapter 5 and Table B.1 define the fourth fuzzy input as resource availability, but later chapters repeatedly call it organizational quality or dataset quality. Use the implemented factor name consistently unless the code is changed and the benchmark is rerun. Also correct the prose around Table 7.2: the 4.10/0.88 item is metadata-supported decision making, not the separate relevance-judgment item shown as 4.40/0.70.
- Step 8. **Describe the baselines exactly and discuss the simplest competitor honestly.** The so-called metadata-quality baseline is a linear weighted score that includes portal-position relevance as well as recency, completeness, and resources; it is not metadata-only. Name it as a linear multi-criteria or weighted-sum baseline and report the implemented weights. It descriptively outperforms fuzzy on the aggregate

metrics and is itself readily interpretable. The discussion must therefore explain what fuzzy rule-based reasoning adds beyond a simpler interpretable weighted sum, and must not imply that fuzzy logic is the only route to transparent ranking.

- Step 9. **Recalibrate the Abstract and Chapter 8 only after the corrected rerun.** Replace 'successfully supports', 'contributes meaningfully', 'demonstrates', and 'improved' where the design supplies only descriptive or perceptual evidence. A defensible summary would report competitive point estimates relative to lexical approaches, explicitly state which adjusted comparisons are or are not significant, identify explanation transparency as the design contribution, and report positive formative perceptions from $n=10$. Do not claim that explanations caused greater trust or that fuzzy ranking improved retrieval unless a controlled comparison and the corrected statistics support those claims.
- Step 10. **Expand the limitations to match the actual design.** Add the single-assessor and author-bias limitation; the small 15-query benchmark; portal-generated candidate-set bias and the absence of end-to-end recall measurement; the single portal and 500-dataset scope; hand-designed rules; $n=10$ and a technically homogeneous sample; the single-condition questionnaire; and the inability to infer that explanations caused changes in understanding or trust. Publishing the complete data and pipeline is a mitigation, not a reason to omit these limitations.

Report the analyses already computed

- Step 11. **Fix 'Table ??' and insert the per-query results.** Section 7.3.2 currently promises a table that is absent. Include the complete table from the corrected pipeline or a readable $nDCG@10$ -only version with 15 queries by five systems. Name specific comparisons rather than declaring an overall winner: for example, fuzzy is strong on MOB-01 and improves over several baselines on ENV-04, while it loses clearly on ENV-02, MOB-04, and parts of XD-05. Explain the query or metadata characteristics that plausibly produce those patterns, without turning post-hoc interpretation into causal proof.
- Step 12. **Discuss ENV-01 as a corpus-coverage result.** All systems score zero because the candidate corpus contains no relevant air-quality dataset. Keep the query because it exposes a real limitation, but state that it reflects coverage or candidate-generation failure rather than a ranking distinction and lowers every system's average. Consider also reporting a sensitivity check with ENV-01 excluded, clearly labelled as supplementary rather than replacing the primary benchmark.
- Step 13. **Insert a compact significance table.** After fixing and rerunning the pipeline, show one headline metric with system pair, Wilcoxon statistic, uncorrected p-value, Holm-corrected p-value, and significance decision. Explain the correction family. In the currently committed results, fuzzy is not significantly different from Portal, Metadata, BM25, or Semantic after correction; only Portal/Metadata versus Semantic reach significance for MAP and $nDCG@10$. Do not preserve these exact sentences if the real semantic baseline changes the results.
- Step 14. **Write the missing sensitivity subsection using the same production configuration.** Add Section 7.3.4 with the rule-weight perturbation table. First resolve why the sensitivity baseline reports MAP 0.8191 while Table 7.1 reports 0.7839 for Fuzzy; this usually means the sensitivity runner is not using the same ranker or candidate configuration. Re-run it through the identical frozen production pipeline. The current files test rule-weight changes only, whereas Chapters 6 and 7 promise membership-function breakpoint perturbations and leave-one-rule-out analysis. Perform the promised analyses or delete those promises.

- Step 15. **Delete the phantom heuristic evaluation.** The roadmap at the end of Section 7.2 still announces a heuristic evaluation section that does not exist. Delete the phrase unless a genuine method and result are added.

Tell the methodology and explanation story exactly as it happened

- Step 16. **Rewrite every section that claims two independent judges.** The incorrect protocol appears not only in Sections 6.2 and 6.8 but also in Section 6.5. Remove the second judge, adjudication discussion, Cohen's kappa, and percentage agreement where they did not occur. Rewrite Sections 6.2, 6.5, 6.8, 6.9, 7.9, and any summary that depends on that story. Remove the Cohen reference if it is no longer used. Describe the real sequence: pooled candidates from all configured systems, author-assigned grades, a later consolidation/finalization pass, and publication of the artifacts for inspection.

Suggested direction (rewrite in your own words): Relevance judgments were assigned by the author as a single assessor to the pooled candidates returned by the evaluated systems. The labels were reviewed in a separate consolidation pass before the final judgment file was used for evaluation. Single-assessor judgments cannot be checked for inter-rater consistency and may favour the proposed system; the complete files are therefore published for audit.

- Step 17. **Make Appendix C a faithful, fully traced explanation.** Replace 'closely matched the query terms' with one actual output from the corrected system. Give the query, dataset identifier and title, matched terms and metadata fields, exact update date or fixed-reference recency value, resource count, completeness value, activated fuzzy rules and strengths, defuzzified score, and the final explanation shown to the user. Derive every value from a saved trace; do not invent an attractive example.

Consistency and presentation

- Step 18. **Complete the visible correction sweep.** Delete the duplicated opening paragraph of Chapter 6. Fix '500 datasetsg'. Reconcile 'Bachelor's/Master's students' with the actual participant group. Resolve Table ?? and the visible [? 5] and [? 6] citations. Correct missing spaces before citations, repair acronym definitions and first uses, and verify that every figure and table is introduced and discussed in the text. Recheck the page containing Holm-Bonferroni and all pages affected by newly inserted tables after the final compile.
- Step 19. **Reconcile front matter, appendices, and institutional metadata.** The thesis title page names the University of Neuchâtel while the README names the University of Fribourg/Human-IST. Use the official institutional wording and template consistently. Check that Appendix A matches Table 7.2, Appendix B matches the implemented membership functions and factor names, and table headers such as 'Variable' and 'Linguistic Term' render without stray spacing. Update the contents page after all new subsections are inserted.

Repository and submission gate

Step 20. **Rewrite the README to describe the completed study, not the original plan.**

Use the same research questions as the thesis; replace '15-20 participants' and 'within-subjects design' with the actual $n=10$ formative design; list the actual five systems and the correct semantic implementation; fix the placeholder clone URL; point tests to the real directory; and make the project tree match the default branch. Remove dead entry points and duplicate prototype paths. Pin dependency versions, including the embedding model revision, and document the supported Python version.

Step 21. **Provide one fresh-clone command that reproduces Table 7.1 exactly.** The command must use the committed snapshot, fixed evaluation date, final single-assessor judgments, real semantic baseline, and production fuzzy ranker without live network access or manual file copying. Record checksums for the snapshot, judgment file, and generated summary. Fail loudly if an authoritative input is missing. Run the command in a clean environment, run the tests, and compare the regenerated CSV byte-for-byte or numerically to the published Table 7.1 values.

Step 22. **Perform the final PDF gate.** Compile twice so references and the table of contents settle. Search the PDF for `??`, `[ ?`, 'two independent', 'kappa', 'adjudicated', 'mock', '500 datasetsg', 'highest MRR', 'organizational quality', and the old participant wording. Inspect every page at 100% zoom for overfull lines, clipped tables, broken citations, orphaned headings, and unreadable figures. Finally, verify that the thesis PDF, repository default branch, and archived evaluation outputs all describe the same experiment.

Do not submit until all of these are true

- The semantic baseline is real and pinned, or it and all related claims have been removed.
- The benchmark runs from a frozen snapshot with a fixed date and clearly states its reranking scope.
- The public annotation files no longer imply two independent judges, and the final relevance scale is explicit.
- The exact questionnaire and auditable aggregates support every value in Table 7.2.
- The Wilcoxon statistics and confidence intervals are computed and labelled correctly.
- The same four research questions appear in Chapters 1, 7, 8, the Abstract, and the README.
- No sentence says Fuzzy has the highest MRR or uses organizational quality unless the corrected results/code support it.
- The per-query, significance, and sensitivity results are present and use the corrected production configuration.
- A fresh clone reproduces Table 7.1 with one documented command and no live-data drift.
- The final PDF contains no unresolved references, duplicated paragraphs, visible typos, or layout defects.